

MATTHEW FOOTE

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Mechanical engineering student who has been building and maintaining machines since age 13 — from a self-built 3D printer, to the eight-printer prototyping lab I kept running at Enlisted Design, to a liquid ethanol rocket engine in progress. Comfortable keeping equipment running, teaching others the tools, and keeping a shop clean and organized. Seeking the [student lab assistant] position in the BYU Mechanical Engineering lab.

EDUCATION

Brigham Young University Provo, UT <i>B.S., Mechanical Engineering</i>	Expected [Month Year]
First-year student, started Fall 2026. Planned involvement: BYU Rocketry Club (propulsion team). [Add GPA / coursework if desired.]	
Lone Peak High School Highland, UT	Aug 2021 – Apr 2024

LAB & SHOP SKILLS

3D printing: Built my first FDM printer at 13. Fleet maintenance and repair: hot-end cleaning and replacement, belt installation, motherboard and firmware upgrades, print troubleshooting and tuning.

Fabrication: Welding and woodworking (formal coursework), CNC, power and hand tools, carbon fiber mold-making and layup, custom camera rigs and cranes.

CAD & software: Onshape, Fusion 360; Bambu Lab and Prusa slicers; DaVinci Resolve, Premiere Pro; Jira; HTML/CSS. Fluent in Tagalog.

EXPERIENCE

Content Strategy Intern — Enlisted Design Salt Lake City, UT	May 2022 – Aug 2022
- Kept the studio's prototyping lab running: eight 3D printers producing models for the industrial design team — cleaned and replaced hot ends, installed belts, upgraded motherboards, and fixed whatever failed. - Documented the design process on one of the studio's largest projects to date, setting up and shooting engineer interviews; footage was used and shared by the team. - Manager review cited maturity, professionalism, and attention to detail; work exceeded expectations.	
Content Manager & Director of Photography — Phuture Films Salt Lake City, UT (part-time)	Sep 2022 – Apr 2024
- Built and led a production team that made five award-winning films in a short window, winning two national film competitions (Student Television Network). - Managed content and ran full-production shoots; designed and built custom camera rigs and cranes for shots on tight budgets.	
Founder & CEO — North Media House Utah	May 2024 – Sep 2024
Started a design collective making films and ads for brands; built the team, office, and systems, managed client relationships, and drove 110,000+ impressions for clients in the first two weeks.	
Full-Time Missionary — The Church of Jesus Christ of Latter-day Saints Philippines San Pablo Mission	Sep 2024 – Aug 2026
Two years of full-time volunteer service: 60+ hour weeks on an assigned schedule; trained and coordinated daily work for rotating companions across multiple assignments; worked daily in Tagalog.	

ENGINEERING PROJECTS

Liquid Ethanol Rocket Engine Independent • Onshape	2026 – Present
Designing a liquid ethanol/LOX engine: parametric thrust chamber, injector with integrated torch igniter, LOX and ethanol feed adapters, service flanges, and a gimbal stack for future thrust vector control. Previously designed two liquid engines; launches rockets regularly.	
Manta — Dual-Mode XC Handlebar & Carbon Fiber Components Independent	2024 – Present
Designed a parametric handlebar in Fusion 360 from requirements observed while racing; printed and assembled a 1:1 FDM prototype for fit testing. Now building carbon fiber molds and laying up carbon handlebars.	
Compact Electronic Braking System Independent	2026 – Present
Developing a compact electronic braking system for bicycles; in design and prototyping.	
Race-Bike Aero Fairings • DIY 3D Printer Independent	2024 / age 13
Modeled and printed aero fairings for XC race bikes; at 13, built a 3D printer and designed its housing in Fusion 360.	

ACTIVITIES

Competitive XC mountain bike racing (NICA varsity) • Ultrarunning — 50-mile run, Aug 2024 • FPV drones • Amateur rocketry